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Defect-dependent environmental stability of high mobility transparent conducting In-doped CdO

Shan Wu[1], Shen Jie Zha[1], Yang Zhang[1], Gui Shan Liu[1], Xiong Jing Chen[1], Zhan Hua Li[2], Chun Yuen Ho[3], Bei Deng[1], Kin Man Yu[4], Chao Ping Liu[1] *

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ABSTRACT

Highly degenerate n-type CdO with high electron mobility is a promising transparent conducting oxide (TCO) for optoelectronic devices utilizing spectrum in the Vis-NIR range. In particular, it has been shown that doped CdO thin films can show much superior transparency of >80% in the NIR region compared to conventional transparent conducting oxide (e.g., Sn-doped In_2O_3) thin films with similar sheet resistance. However, CdO thin films typically experience rapid degradation in their electron mobilities when exposed to environmental conditions with H_2O moisture. Here, we studied the effects of thermal annealing on the environmental stability of In doped CdO (CdO:In) using a combination of different analytical techniques. CdO:In thin films with different In concentration (0~8.3%) synthesized by magnetron sputtering were subjected to different post thermal annealing (PTA) and then aged in different environmental conditions with varying relative humidity (RH) in the range of 0~85%. Our results reveal that the degradation of CdO:In thin films can be primarily attributed to the oxygen vacancy-related defects at the grain boundaries which can readily react with the OH^- in the moisture. The moisture induced degradation can be mitigated by appropriate PTA at high temperature (>400 °C) where grain boundary defects, primarily associated with Cd vacancies, can be passivated through hydrogen (H), and thus enhancing their environmental stability. The present study provides a comprehensive understanding of the instability mechanisms and defect passivation in transparent conducting CdO:In thin films, which can also be relevant for other wide gap oxides.

I. INTRODUCTION

Transparent conducting oxides (TCOs) have attracted significant attention in recent decades due to their widespread applications in various optoelectronic devices, ranging from solar cells to nonlinear optics[1][2][3][4][5][6][7][8][9][10][11][12]. Among them, In_2O_3-based TCOs such as In_2O_3:Sn (ITO) dominate the current TCO market owing to their exceptional optoelectronic performance, including high conductivity ($\sigma \sim 10^4$ S/cm) and high transparency (~85%) in the wavelength range of 300–1000 nm [2]. However, conventional TCOs like ITO, doped SnO_2 or ZnO often suffer from substantial free carrier absorption and plasma reflection at longer wavelengths (λ >1000 nm), thus limiting their effectiveness in applications that utilize the near infrared (NIR) spectrum, such as multi-junction solar cell, infrared photodetectors [1][13][14]. To address this limitation, alternative TCO materials with lower free carrier absorption coefficient (α_{FCA}) and a longer plasma wavelength (λ_p) have been explored. According to the Drude model [13][15], the free carrier absorption coefficient α_{FCA} is influenced by the free carrier density N and the carrier mobility μ,

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i.e., $\alpha_{\mathrm{FCA}} \propto N/\mu$. Consequently, in order to achieve a high conductivity while minimizing α_{FCA}, TCOs with relatively high μ and moderate N are desirable.

Among TCOs, CdO stands out with the highest electron mobility (μ) of approximately 100–200 $cm^2V^{-1}s^{-1}$, even at an N of 10^{20}–10^{21} cm^{-3}, so that it can achieve a high conductivity (σ) of over 10^4 S/cm [16][17][18][19][20]. The remarkable μ of CdO can be attributed to its Cd 5s-based conduction band, which results in a low electron effective mass, and its large static dielectric constant (ε_s=21.9), effectively screening ionized donors and minimizing ionized impurities scattering [17]. Nominally "undoped" CdO thin films can exhibit a high N ranging from 1×10^{20} to 4×10^{20} cm^{-3}, primarily due the presence of oxygen vacancies (V_O) and H impurities (primarily in form of H_O) [19]. Furthermore, appropriate extrinsic doping with various dopants, such as In, Ga, V, Ti, can increase the N of CdO to exceed 10^{21} cm^{-3} [8][17][20]. In addition to doping, alloying CdO with other (ultra)wide bandgap semiconductors (e.g., ZnO, In_2O_3, Ga_2O_3, NiO) can achieve transparent semiconductors/conductors with widely tunable electronic bands[21][22][23][24][25][26][27]. Among dopants commonly used to increase the N of CdO, In has been found to be particularly effective in achieving N values exceeding 10^{21} cm^{-3} while still maintaining μ>~100 $cm^2V^{-1}s^{-1}$ [16][20]. While the intrinsic direct bandgap (E_g) of CdO is approximately 2.3 eV, the optical bandgap (E_g^{opt}) of heavily doped CdO can be widened to around 3.2 eV due to the Burstein-Moss (BM) shift. Additionally, the high-frequency dielectric constant (ε_∞) of CdO is larger than that of conventional TCOs such as ITO or FTO [13], resulting in a longer plasma wavelength λ_p >1200 nm even with N>10^{21} cm^{-3}. Hence, degenerately doped CdO is a promising TCO material with relatively high transparency across the UV to NIR spectral range.

However, most technologically important TCOs (e.g., Al-doped ZnO, doped CdO, H-doped In_2O_3) typically suffer from insufficient stability when exposed to a humid environment[28][29][30][31], largely limiting their wide-ranging applications. Therefore, investigating the environment stability as well as degradation mechanisms of these promising TCO materials is crucial for their practical utilization. For instance, Walukiewicz and co-workers discovered that undoped CdO with a positively charged surface decomposes rapidly in a humid environment (85 ºC/85% RH), whereas heavily n-type doped CdO with a high density of negative surface charge exhibits remarkable stability, primarily governed by the sign and density of surface charge [30] due to the high position of the Fermi level. Ballif et al. observed significant degradation in the electron mobility of H-doped In_2O_3 after damp heat treatment, mainly caused by chemisorbed OH^- or H_2O-related species [31]. The degraded electrical properties of sputter-deposited Al-doped ZnO (AZO) thin films after exposure to a damp heat environment can be restored by appropriate annealing [28]. Nevertheless, a consensus on the degradation mechanisms of TCO materials is currently still lacking.

In this study, we aim at exploring the environmental stability of In-doped CdO (CdO:In) thin films and the corresponding instability mechanisms. CdO:In thin films with different In concentration (0~8.3%) were synthesized by magnetron sputtering and were subjected to aging in different environmental conditions with varying relative humidity (RH) in range of 0~85%. We employed a combination of analytical techniques including Hall measurement, spectroscopic ellipsometry (SE) and X-ray photoelectron spectroscopy (XPS) to study the optoelectronic properties. Our results show that the mobility degradation of CdO:In thin films is primarily attributed to the oxygen vacancy-related defects at the grain boundaries which can readily react with the OH^- when exposed to moisture. We further demonstrate that the moisture induced degradation can

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be alleviated by appropriate PTA at high temperature (>400 ºC) where grain boundary defects are passivated by hydrogen (H).

II. EXPERIMENTAL METHODS

The CdO:In thin films were grown on glass substrates using a multi-gun radio frequency magnetron sputtering system. The system includes separate CdO and In_2O_3 targets, and the deposition was carried out at an elevated substrate temperature of ~300 ºC. The In concentration (In%) in the films was controlled within the range of 0~8.3% by adjusting the sputtering power of the In_2O_3 target. Prior to sputtering, the chamber was evacuated to a base pressure of ~7.5×10^{-6} Torr. The sputtering process was conducted using pure Ar gas at a working pressure maintained at ~3.7 mTorr. The thickness of the thin films is in range of 80~100 nm. Post thermal annealing (PTA) treatments were performed at different temperatures (T_A) between 300 and 500 ºC for varying durations of 5~10 minutes in either ambient air or an Ar atmosphere. Environmental stability testing was performed under various environmental conditions at room temperature (ranging from 27 to 35 ºC) with different relative humidity (RH) levels, i.e., pure Ar with ~0% RH, air with different RH (i.e., ~1% RH, ~45% RH, ~85% RH). The In concentration (i.e., In%=[In]/{[Cd]+[In]}) was determined using energy-dispersive X-ray spectroscopy (EDS). The crystal structures of thin films were analyzed using grazing incidence x-ray diffraction (GIXRD) with a grazing incidence angle of 1°. Spectroscopic ellipsometry (SE) measurements were performed at room temperature in the spectral range of 0.74-5.9 eV to study the optical properties of the films. The electrical properties were characterized by room temperature Hall-effect measurements in the van der Pauw configuration. High-resolution XPS measurements were conducted using a monochromatic Al Kα x-ray source (hν=1.487 keV), and the photoelectrons were collected and analyzed using a concentric hemispherical analyzer.

III. RESULTS AND DISCUSSION

3.1 Structural properties

Fig. 1(a) displays the GIXRD patterns of as-grown CdO:In thin films with varying In concentration ranging from 0 ~8.3%. As shown, these CdO:In thin films are polycrystalline with the rock salt (111) and (200) diffraction peaks, consistent with previous reports[27][32]. The diffraction peaks of these CdO:In thin films with different In% do not exhibit noticeable shift compared to CdO, indicting their similar lattice parameters. This is reasonable since the ionic radii of Cd^{2+} (1.09 Å) and In^{3+} (0.94 Å) are relatively close [20]. By applying the Scherrer equation, the grain sizes of these as-grown CdO:In thin films are estimated to be in range of 18~19 nm. Comparatively, the GIXRD patterns of the CdO:In thin films subjected to PTA treatments (e.g., at T_A of 400~500 ºC for 10 minutes) in ambient air or Ar show a slight decrease in the corresponding full width at half maximum (FWHM) values (not shown). Consequently, the grain sizes of the CdO:In thin films after PTA treatments are found to be in range of 19~20 nm, which indicates that grain growth is not significant at relatively low temperature and short annealing time.

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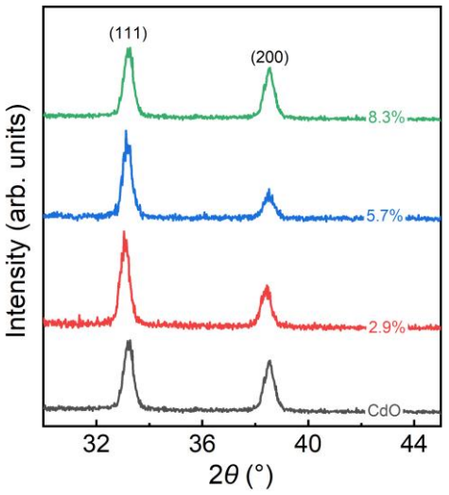


Fig. 1. GIXRD patterns of as-grown CdO:In thin films on glass substrates with varying In% ranging from 0 to 8.3%.

3.2 Electrical properties

The electrical properties (electron concentration N, electron mobility μ, resistivity ρ) of as-grown CdO:In thin films (denoted as black squares) with different In% are shown in Fig. 2. With increasing In%, the N of as-grown CdO:In thin film increases linearly from ~1×10^{20} cm^{-3} for the CdO to ~1.2×10^{21} cm^{-3} for the CdO:In with an In% of ~8.3% (top panel of Fig.2a). Meanwhile, the corresponding μ is rather stable and varies only slightly in the range of 80~108 $cm^2V^{-1}s^{-1}$ (middle panel of Fig.2a), resulting in a monotonic decrease in the resistivity with In% from ~5.2×10^{-4} Ω cm to ~4.9×10^{-5} Ω cm (bottom panel of Fig.2a). The μ of the CdO:In thin film depends on various scattering mechanisms, such as phonon scattering, ionized impurity scattering, grain boundary scattering [33]. Temperature dependent Hall-effect measurements in the temperature range of 80 to 300 K (data not shown) show that the μ of CdO:In only varies slightly on temperature, suggesting that the mobility in these highly degenerate CdO:In thin films is primarily limited by temperature-independent defect scattering with a small contribution from phonon scattering. For instance, the CdO:In (~8.3%) shows a small decrease of μ as the temperature increases from 80 to 300 K, with a $\Delta\mu/\Delta T\approx$-0.088 $cm^2V^{-1}s^{-1}K^{-1}$.

These as-grown films were then subjected to PTA treatment in either Ar or ambient air (at 400 ºC for 10 minutes) and their electrical properties are also illustrated in Fig. 2(a). It can be observed that the annealing environment strong affects the electrical properties of the samples. When PTA in ambient air (blue triangles), the N exhibits a notable reduction with $|\Delta N|/N$ in the range of 18%~58%, with larger $|\Delta N|/N$ for samples with higher In%. On the other hand, when subjected to PTA in Ar (red circles), films with In% up to ~6% exhibit only minor change in N ($|\Delta N|/N$ <4%). However, the N for the film with high In% of 8.3% drops significantly by more than a factor of 2 from 1.2×10^{21} to $5\times10^{20}cm^{-3}$. The significant reduction in N for the CdO:In after PTA with high In% of ~8% (in Ar) and >6% (in air) can be mainly attributed to the dopant segregation, which is more likely in an O_2 environment (air) which favors the removal of In from the Cd site forming In_2O_3 clusters.

Interestingly, the μ of CdO significantly increases by ~88% to >160 cm^2/Vs after PTA treatment in air, while it decreases slightly by ~10% after PTA treatment in Ar.

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This can be understood by considering that in undoped CdO thin films, the high electron concentration comes from O-deficient native donors (O vacancies V_O and/or Cd interstitials Cd_i). When annealed in an O-rich environment (air), these native donors are effectively reduced, which gives rise to the lowering of N (by a factor of ~2.4) and increases in μ due to reduced ionized defect scattering. On the other hand, PTA treatment in an Ar environment is much less efficient in removing these native donors, and hence only has a minor effect on the electrical properties. With increasing In%, the μ also changes in accordance with the change in N, i.e. a decrease in N gives rise to an increase in μ, which is in agreement with the removal of active In after PTA for the samples with high In%. Due to the decrease in N upon PTA treatments (in Ar or air), the ρ of these CdO:In thin films increases with a relative change in resistivity $\Delta\rho/\rho$ in the range of 8%~172%.

Fig. 2(b) summarizes the electrical properties (i.e., μ *vs.* N) of freshly prepared CdO:In thin films, including both the as-grown samples and samples with PTA treatments. It can be observed that the μ generally decreases as the N increases. CdO:In thin films with relatively high In% (>5%) exhibit a wide range of changes in their N (e.g., from ~4×10^{20} cm^{-3} to ~12×10^{20} cm^{-3}) upon different PTA treatments (T_A: 300~500 °C; duration:5~10 minutes; atmosphere: Ar or air), due to the large variations in dopant segregation and the formation of compensating acceptors. For typical polycrystalline TCO thin films, the μ may also be limited by the grain barrier scattering (GBS), which is related to the potential barrier height Φ_b. The barrier height can be expressed as $\Phi_b = \frac{e^2 N_t^2}{8\varepsilon_s\varepsilon_0 N}$, where ε_s is the static dielectric constant, ε_0 is the permittivity of vacuum, N_t is the charge carrier trap density at boundaries and N is the free carrier concentration [33]. In contrast to the strong GBS as observed in Al-doped ZnO thin films with N<2×10^{20} cm^{-3} [34], the large static dielectric constant ε_s of CdO can effectively diminish GBS, allowing for achieving high μ even in case of relatively low N (~10^{19} cm^{-3}) in polycrystalline films. In other words, comparing to other TCOs, the mobility of CdO is much less sensitive to grain size. Fig. 2(b) also shows that TCOs with conductivity >10000 S/cm can be readily achieved with CdO:In thin films.

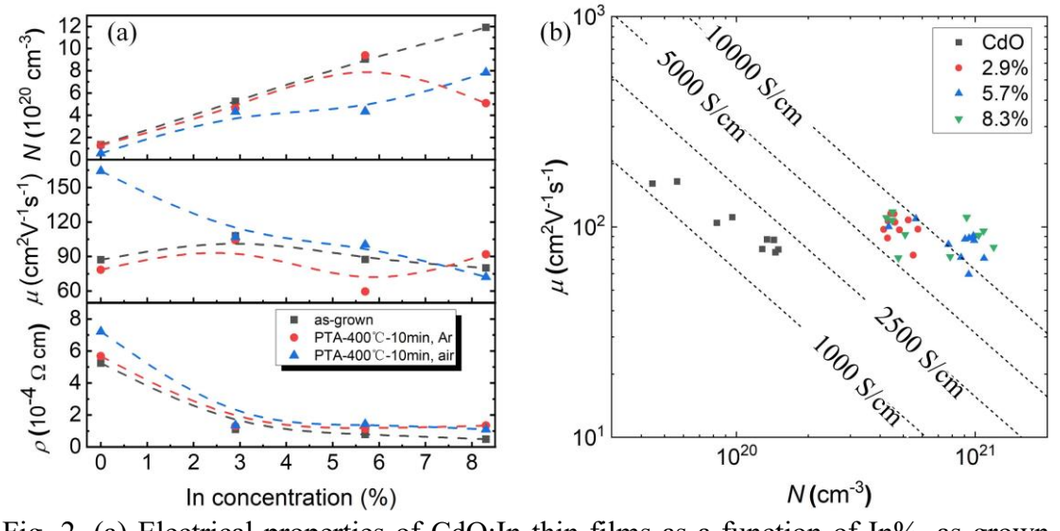


Fig. 2. (a) Electrical properties of CdO:In thin films as a function of In%, as grown (black squares), PTA at 400 °C for 10 minutes in Ar (red circles) and in air (blue triangles). Dashed lines are the guilds to the eyes. (b) A summary of measured μ for these CdO:In thin films with varying In% after undergoing different treatments,

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including as-grown samples and PTA at T_A ranging from 300-500 °C for durations of 5-10 minutes in either Ar or air. The dashed lines in the figure indicate constant conductivities.

3.3 Optical properties

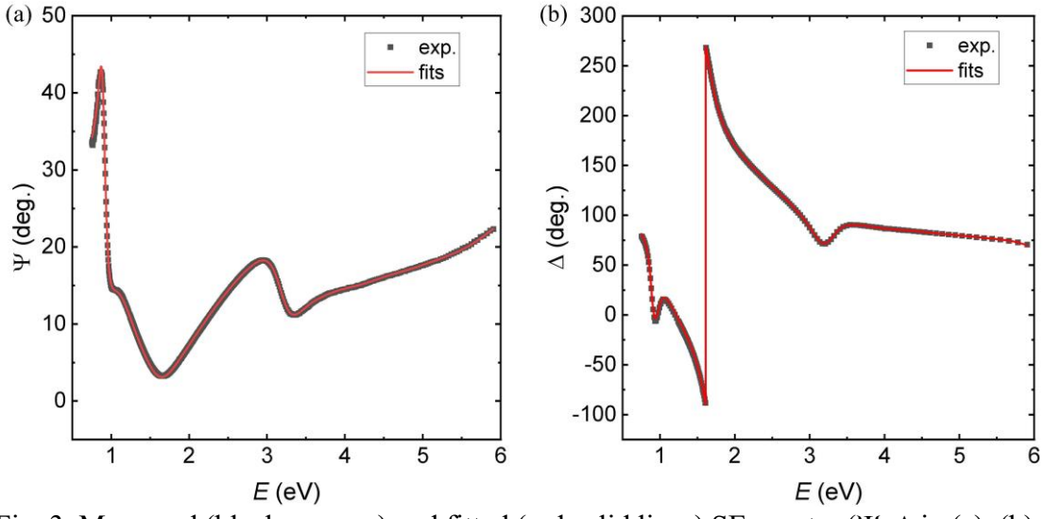
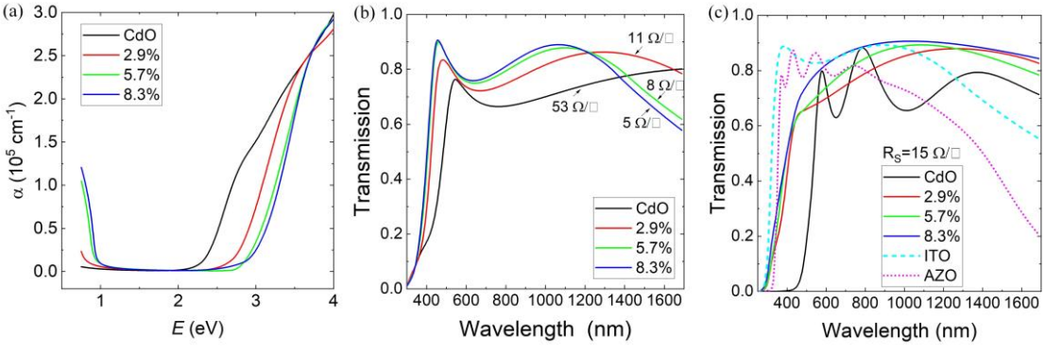


Fig. 3. Measured (black squares) and fitted (red solid lines) SE spectra (Ψ, Δ in (a), (b), respectively) as a function of photon energy E for a typical as-grown CdO:In thin film (In%= 5.7%), with angle of incidence of 65°.

The optical properties of these thin films were studied using SE. SE data analysis is based on a three-layer optical model, consisting of a glass substrate, the CdO:In thin film, and a surface roughness layer. To describe the dielectric functions of the CdO:In thin films, a Tauc-Lorentz oscillator, 2~3 Gaussian oscillators and a Drude oscillator are employed. The dielectric function of the surface roughness layer is modeled as a 50/50-vol% mixture of the CdO:In thin film and voids [35]. The detailed parameters in these oscillators as well as the data fitting/analysis procedure can be found in the literatures [13]. In SE measurements, the change in the polarization state of the incident light is described by the ellipsometric Ψ(amplitude ratio) and the Δ (phase difference), given by r_p/r_s=tan (Ψ)exp(iΔ), with r_p and r_s the reflection coefficients for p-polarized light and s-polarized light respectively. Fig. 3 displays the SE spectra (both measured and fitted values with an angle of incidence of 65°) as a function of photon energy (E) for a representative as-grown CdO:In thin film (with In%=5.7%). Fig. 3 shows that these dielectric function models can fit the experimental SE data (Ψ, Δ) very well over the whole spectral range with a mean square error (MSE) of approximately 8.0.



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Fig.4 (a) Absorption coefficient α as a function of photon energy E for the as-grown CdO:In thin films with varying In%; (b) transmission spectra of CdO:In thin films calculated from SE results for a film thickness of ~100 nm on glass substrates, with their corresponding Rs indicated; (c) transmission spectra of CdO:In thin films with varying In% on glass substrates with film thickness calculated for a Rs~15 Ω/□, compared to transmission spectra of ITO and AZO thin films with the same Rs~15 Ω/□.

Based on the SE analysis, the dielectric function of the thin film, denoted as $\varepsilon(E) = \varepsilon_1(E) + i\varepsilon_2(E)$, can be derived, where the $\varepsilon_1(E)$ and the $\varepsilon_2(E)$ are the Kramers-Kronig transforms of each other. The complex refractive index $\tilde{n} = n + i\kappa$ can also be derived from the dielectric function using the Maxwell relation, where n and κ are the refractive index and extinction coefficient, respectively. The optical absorption coefficient (α) at wavelength λ can be calculated using $\alpha = 4\pi\kappa/\lambda$. Fig. 4(a) plots the α as a function of E for the as-grown CdO:In thin films. It can be observed that with an increase in In% content, the optical absorption edge undergoes a blue shift, while the free carrier absorption in the NIR region significantly increases. Fig. 4(b) displays transmission spectra of these CdO:In thin films calculated from SE results for a film thickness of ~100 nm on glass substrates, along with the corresponding sheet resistance (R_S) indicated. We note that typically, a sheet resistance $R_S < 20$ Ω/□ is desired for transparent electrodes. Fig. 4(c) compares the transparency of CdO:In films and common TCOs (ITO and AZO) with R_S=15 Ω/□. The ITO film shown here has N=1.34×10^{21} cm^{-3}, μ=32.6 cm^{2}V^{-1}s^{-1}, ρ=1.4×10^{-4} Ω cm and AZO has a N=3.8×10^{20} cm^{-3}, μ=24.1 cm^{2}V^{-1}s^{-1}, ρ=6.7×10^{-4} Ω cm. As depicted in Fig. 4(c), the CdO:In thin films exhibit high transmission (ranging from 60% to 90%) in the spectral range of 400~1700 nm. Notably, these CdO:In thin films demonstrate significantly improved transparency in the wavelength range above 1100 nm compared to conventional ITO or AZO thin films. This demonstrates their potential applications in optoelectronic devices that require full spectrum utilization from visible to NIR regions.

The optical band gap E_G^{opt} can be estimated by extrapolating α^2 to zero photon energy. Due to the high N in these films, their E_G^{opt} will be larger than its intrinsic direct band gap (~2.3 eV) due to the effects of Burstein-Moss (BM) shift and band-gap renormalization (BGR). The optical band gap can be expressed as

$$E_G^{opt} = E_G + \Delta E_G^{BM} - \Delta E^{BGR}, \qquad (1)$$

where E_G, ΔE_G^{BM}, ΔE^{BGR} are the intrinsic band gap, the band gap widening by the B-M shift and the band gap shrinkage by the BGR, respectively [36][37][38]. Fig. 5(a) illustrates the E_G^{opt} as a function of N for CdO:In thin films with different In% and various PTA treatments. It is observed that the E_G^{opt} for these CdO:In thin films ranges from 2.4 to 3.1 eV. In Fig. 5(b), the band-gap shift ($E_G^{opt} - E_G$) for these CdO:In thin films is plotted as a function N, assuming E_G=2.3 eV. The ΔE_G^{BM} and ΔE^{BGR} are calculated by using the nonparabolic conduction-band model and the Jain's model, respectively [13]. The calculated ΔE_G^{BM}, $\Delta E^{BGR}$and $\Delta E_G^{BM} - \Delta E^{BGR}$ are also shown in Fig. 5(b) for comparison. It can be observed that the experimentally obtained band-gap shift ($E_G^{opt} - E_G$) of CdO:In thin films agrees well with the calculated values.

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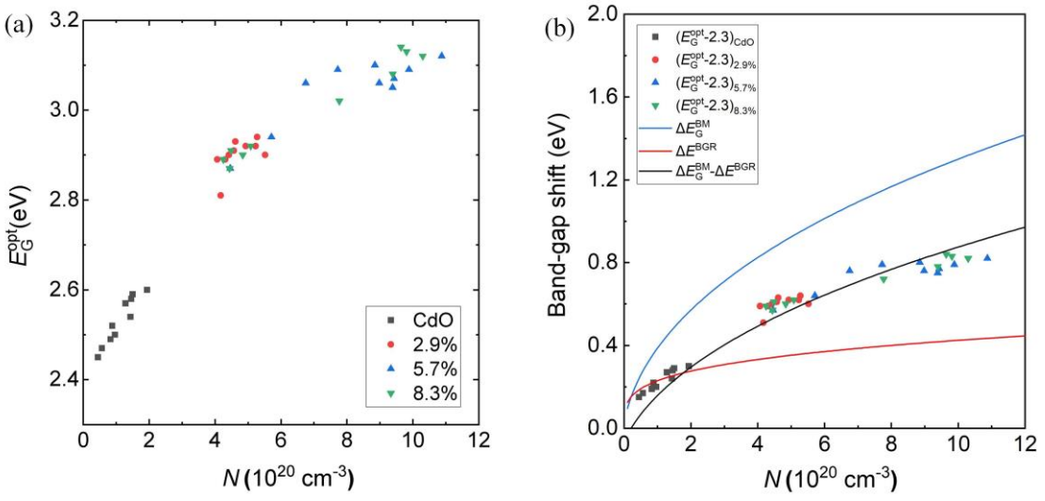


Fig. 5. (a) Optical band gap of CdO:In thin films as a function of N, and (b) band-gap shift of CdO:In films as a function of N, assuming an intrinsic direct band gap of 2.3 eV. The calculated Burstein-Moss shift ΔE_G^{BM}, the band-gap renormalization ΔE^{BGR}, and $\Delta E_G^{BM} - \Delta E^{BGR}$ are also shown.

3.4 Environmental stability

To investigate the environmental stability, these CdO:In thin films were aged under various environmental conditions including pure Ar (~0% RH) and air with three different RH values (i.e., 1% RH, 45% RH, 85% RH) at room temperature (27~35 ºC). Fig. 6(a) shows the electrical properties of CdO:In thin films (after PTA treatment at 300 ºC in Ar for 10 minutes) as a function of aging time (up to 120 days) in ~45% RH. It is worth noting that the CdO:In subjected to PTA treatments at 300 ºC for 10 minutes exhibited similar electrical properties and environment stability to the as-grown samples. While negligible change can be observed in the N (shown in Fig. 7), these films exhibit a distinct reduction in μ during aging in 45% RH. The μ for all films drop by >40% after 20 days in 45% RH and after 120 days by >2-5 times. While the Hall mobility μ is affected by grain boundaries, we also calculated the in-grain optical mobility (μ_{opt}) using the equation $\mu_{opt}=e\tau/m^*$, where e the elementary charge, τ the mean scattering time derived from the Drude model in SE analysis [15], and m^* the electron effective mass [39]. According to the nonparabolic conduction band model [40], the dependence of m^* on the electron concentration N can be described by

$$m^* = m_o^* \sqrt{1 + 2C\frac{\hbar^2}{m_o^*}(3\pi^2 N)^{\frac{2}{3}}} \quad , \qquad (2)$$

where m_o^* is the effective mass at the bottom of the conduction band, C is a nonparabolicity parameter. In our previous study for the In or Ga-doped CdO thin films, m_o^* and C were determined to be approximately 0.13 m_0 and 0.5 eV^{-1}, respectively. Here, m_0 is the free electron mass. In the current work, we calculated the m^* of CdO:In thin film with varying N using Eq.(2) and the same parameter values (m_o^*=0.13 m_0, C=0.5 eV^{-1}) as in our previous study[13]. The corresponding μ_{opt} of CdO:In thin film during aging is shown in Fig. 6(b). Interestingly, unlike the Hall μ, the μ_{opt} is found to be nearly unchanged for CdO:In thin films with varying aging time, indicating that the degradation in the mobility is primarily caused by enhanced electron scattering at grain boundaries. It is important to note that the Hall-effect measures lateral carrier transport

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in the thin film, while SE analysis probes the film vertically. Therefore, the difference between the Hall mobility and the optical mobility (expressed as $|\mu-\mu_{opt}|$) can be attributed to possible anisotropic electron transport in these CdO:In thin films [13]. As mentioned earlier, we also observe that the degraded electrical properties of CdO:In thin films can be recovered through appropriate thermal annealing treatments in Ar or ambient air (e.g., 300 ºC for 10 minutes). For instance, the CdO:In (~2.9%) exhibited a degraded electrical properties (N=4.8×10^{20} cm^{-3}, μ=4.9 cm^{2}V^{-1}s^{-1}, ρ=3.3×10^{-3} Ω cm) after 180 days aging in ~45% RH, which was found to be largely recovered (N=4.8×10^{20} cm^{-3}, μ=96.7 cm^{2}V^{-1}s^{-1}, ρ=1.4×10^{-4} Ω cm) after PTA (300 ºC for 10 minutes) in ambient air. A similar phenomenon was reported by F. Machda *et al*. for AZO thin films, which was attributed to the desorption of chemisorbed OH^- or H_2O-related species [28]. As discussed in the following section, the degradation rate of as-grown CdO:In thin films with different In% decreases as the RH% of environmental conditions decreases. Therefore, the reduction in μ for these as-grown CdO:In thin films is also likely due to chemisorbed OH^- or H_2O-related species at the grain boundaries, which acts as scattering centers for free electrons. To verify this conjecture, we carried out a more systematic investigation on the effects of aging under different RH for CdO:In after PTA. Then, we will discuss in more detail how these chemisorbed OH- may react with Cd atoms with dangling bonds (created by the V_O defects) at the grain boundaries, resulting in the formation of $Cd(OH)_2$.

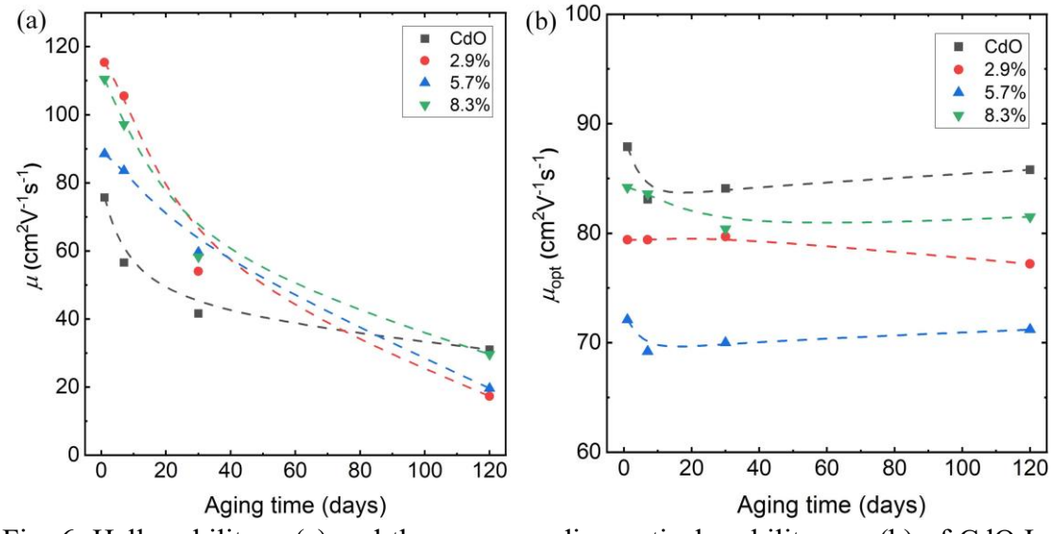


Fig. 6. Hall mobility μ (a) and the corresponding optical mobility μ_{opt} (b) of CdO:In thin films (with PTA at 300 ºC in Ar for 10 minutes) with different In% as a function of aging time in 45% RH.

Fig. 7 shows the normalized electrical properties of three representative CdO:In thin films (i.e., CdO, CdO:In with In% of 2.9%, and 8.3%, shown in Fig7(a), Fig. 7(b) and Fig. 7(c), respectively) as a function of aging time under different environmental conditions. As depicted in the figure, all CdO thin films (undoped and In doped) exhibit excellent stability when aged in Ar (~0% RH), but show significant degradation in their mobility when the RH is greater than 0%, so that their conductivity (σ=1/ρ) decreases with aging time. More importantly, the rate of degradation in μ increases as the RH percentage in the environment increases. For instance, the average degradation rate (i.e., $|\Delta\mu|/\Delta t$) for the CdO:In (2.9%) was found to increase from ~0.13 cm^{2}V^{-1}s^{-1}day^{-1} to

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~3.64 $cm^2V^{-1}s^{-1}day^{-1}$ as increasing of RH from ~0% to ~85%. This observation strongly suggests that the deterioration in the mobility of CdO:In thin films is closely associated with the presence of moisture in the ambient atmosphere, which acts as the source of OH^- or H_2O-related species.

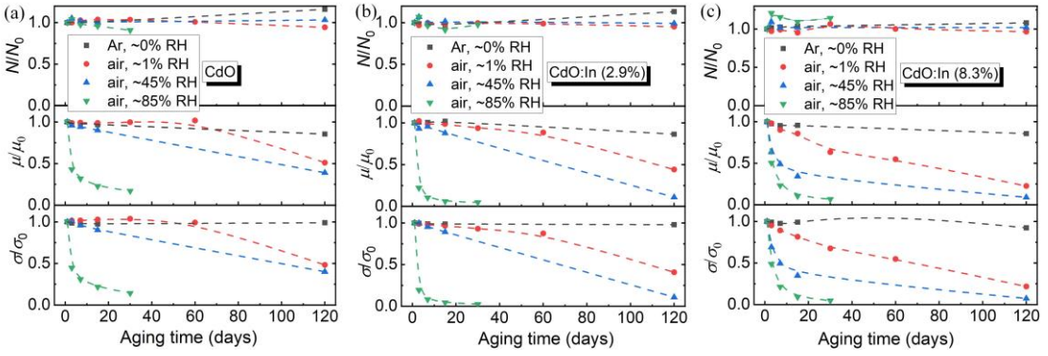


Fig. 7. Normalized electrical properties (from top to bottom panel: electron concentration N/N_0, mobility μ/μ_0, conductivity σ/σ_0, where N_0, μ_0 and σ_0 are the concentration, mobility and conductivity of the unaged sample) of CdO (a), CdO:In (In%=2.9%) (b), and CdO:In (In%=8.3%) (c) thin films (with PTA at 300 ºC in Ar for 10 minutes) as a function of aging time under various environmental conditions with different RH (0~85%).

Defect passivation has been proven effective in improving the optoelectronic properties of transparent conducting oxides [41][42]. In particular, the passivation of defects using PTA treatment has been found to be successful [42][43][44]. To assess the impact of PTA treatments on the environmental stability of CdO:In thin films, aging experiments were conducted in ambient air with ~45% RH. The PTA treatments were performed either in Ar or ambient air at T_A ranging from 400 to 500 ºC for durations of 5~10 minutes. Fig.8(a) displays the normalized electrical properties of CdO thin film as a function of aging time after various PTA treatments. For instance, "PTA-500 ºC-10 min, air" indicates a PTA treatment carried out at T_A of 500 ºC for 10 minutes in ambient air. The figure also includes the electrical properties of as-grown samples for comparison. From the results shown in Fig. 8(a), it is evident that CdO thin films after PTA in air or Ar at 500 ºC exhibit enhanced stability in air with ~45% RH. In particular, for the CdO:In thin films (Fig. 8(b) and Fig. 8(c)), PTA at 500 ºC for a longer duration of 10 minutes results in superior environmental stability with essentially no change in N and μ after 120 days in ~ 45% RH. On the other hand, PTA at a relatively low T_A of 300 ºC for different durations (5~10 minutes) are not effective in improving the environmental stability of CdO:In thin films. However, we note that the electrical properties of these degraded CdO:In thin films can be restored by thermal annealing at low T_A below ~300 ºC. This suggests that moisture related defects which degrade the μ are thermally unstable.

Results shown in Fig. 8 suggests that annealing at temperatures ~400~500 ºC for long duration of 10 minutes can effectively passivates grain boundary defects present in CdO:In thin films. As we mentioned earlier, the predominant defects at grain boundaries of CdO:In thin films, such as V_O-induced dangling bonds on the Cd atoms, can readily react with OH^- ions from adsorbed H_2O, leading to the degradation in the electron mobility. PTA at high temperatures for extended durations can remove these dangling bond defects through H passivation due to the pervasive presence of H (e.g. from H_2O-related molecular species) during growth and processing [44]. A more

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detailed discussion on the degradation mechanisms and the H passivation of grain boundary defects will be discussed in the following section.

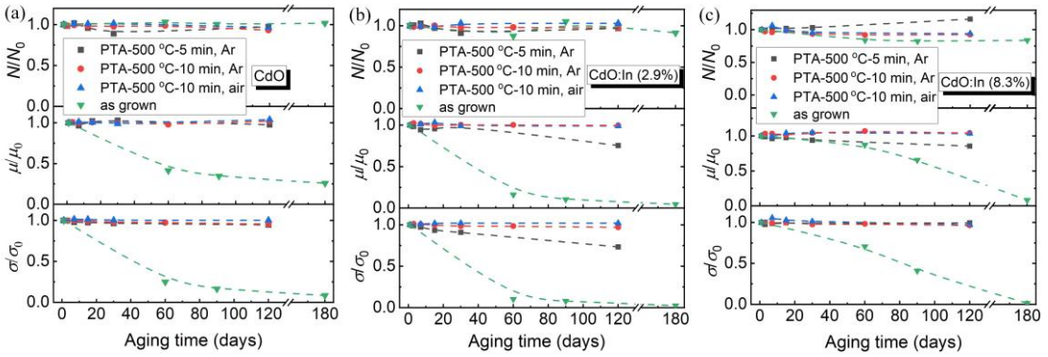


Fig. 8. Normalized electrical properties (from top to bottom panel: electron concentration N/N_0, mobility μ/μ_0, conductivity σ/σ_0) of CdO:In thin films (with different In% (0, 2.9%, 8.3% in (a), (b), and (c), respectively) under various PTA conditions as a function of aging time upon exposure to air with ~45% RH. The electrical properties of the corresponding as-grown (unannealed) samples are also provided for comparison.

3.5 Mobility degradation mechanism and defect passivation

In order to gain a better understanding of the mechanism leading to the environmental instability of CdO:In, XPS measurements were conducted on six representative CdO:In thin films, namely, undoped CdO, CdO:In with 8.3% In after different PTA conditions (300 ºC or 500 ºC for 10 minutes in Ar) and aging durations (1 day or 120 days) in air with ~45% RH. These six samples are listed in Table I as S1-S6. We note again that the samples subjected to PTA treatments at 300 ºC for 10 minutes exhibited similar electrical properties and environment stability to the as-grown samples. All XPS spectra are recorded after a brief Ar^+ sputtering to remove possible surface contaminations. The binding energy (BE) are referenced to the C 1s peak at 284.8 eV. Table I also lists the μ/μ_0 of the samples which shows that significant degradation for sample S2: CdO PTA at 300 ºC with an aging time of 120 days (referred to as "CdO-300 ºC-D120") and S5: CdO:In with 8.3% In after PTA at 300 ºC with an aging time of 120 days ("8.3%-300 ºC-D120"). On the other hand, other samples which are either PTA at 500 ºC (S3 and S6) or only 1 day exposure to ~45% RH (S1 and S4) do not show degradation in their μ. The O 1s core-level spectra for CdO thin films (S1-S3) and CdO:In thin films (S4-S6) are shown in Fig. 9(a) and Fig. 9(b), respectively. The broad O 1s peaks can be deconvoluted into 3 peaks: O_I, O_{II} and O_{III}. The O_I peak at BE ~529 eV can be attributed to O^{2-} in the CdO anion lattice, while the O_{II} peak at BE~530 eV has been associated with oxygen vacancy (V_O) related states [45]. The O_{III} peaks at BE~531.5 eV might be associated with strongly bound oxygen (e.g., OH), loosely bound oxygen (e.g., $Cd(OH)_2$) and even cadmium carbonate (i.e., $CdCO_3$) on the film surface [45][46][47]. Nevertheless, we also note that the assignment of the O_{II} peak is still controversial. For instance, some recent studies suggested that this O_{II} peak most likely results from the adventitious hydroxyls (i.e., OH^-) of water [48] or from water molecules strongly bound to the oxide surface or surface oxygen passivated with hydrogen of water [49] or even from the CdO_2 [50], instead of the V_O related defects. It is worth mentioning that some of these species (e.g., OH^- from water) may readily react with the V_O due to electrostatic attraction of OH^- ions by the positive ionized V_O^+ and/or V_O^{2+}, resulting in the formation of $Cd(OH)_2$. In these cases, the quantity of

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$Cd(OH)_2$ would be directly related with the concentration of V_O in the film. Overall, the presence of V_O in the films may cause two possible differences that can result in the O_{II} spectral shoulder. In the presence of V_O, the local environment of neighboring oxygen, and hence their BE would be different. Moreover, the V_O may react with the OH^- in the moisture, leading to the formation of $Cd(OH)_2$. In other words, the O_{II} peak can be mainly ascribed to the oxygen in the vicinity of the oxygen vacancies[51], and its intensity can be related to the concentration of V_O. Therefore, the area of the O_{II} peak can be used to quantify the concentration of V_O in these films, and thus is also an indication of the density of dangling bonds on the Cd atoms, since for samples without passivation, Cd dangling bonds would be proportional to the concentration of V_O. The peak area ratios O_{II}/O_I and O_{III}/O_I for the samples (S1-S6) are tabulated in Table 1. Interestingly, the O_{II}/O_I peak area ratio is higher for the unstable samples (S1, S2, S4, S5) compared to the stable ones (S3, S6). The elemental ratios of {[Cd]+[In]}/[O] obtained from XPS measurements for these samples are also presented in Table 1. It is observed that {[Cd]+[In]}/[O] values for the unstable samples (S1, S4, S5) are greater than one, further confirming the oxygen deficiency nature and the presence of high density of V_O in these samples. It should be noted that the degraded sample S2 has a {[Cd]+[In]}/[O] value of less than one, likely due to the additional oxygen introduced through the formation of $Cd(OH)_2$ after aging for 120 days. For the stable samples (S3, S6), as expected, their corresponding {[Cd]+[In]}/[O] values are found to be less than one, indicating that PTA at relatively high temperatures (e.g. 500 ºC) is effective in removing O vacancies in these samples, and consequently results in less Cd dangling bonds. In addition, the peak area ratio of O_{III}/O_I for the aged samples (e.g., S2, S3, S5, S6) is higher than that of fresh prepared ones (e.g., S1, S4). This indicates that the concentration of defects or contaminants on the film surface is generally higher after longer exposure in ambient air, and the mobility degradation as observed in the unstable samples mainly results from the scattering by the defects at the grain boundaries rather than the film surface.

Table 1. Summary of peak area ratios of O_{II}/O_I, O_{III}/O_I, and the {[Cd]+[In]}/[O] obtained from XPS analysis for the six samples (S1-S6). The relative mobility μ/μ_o is also given.

Sample ID	Sample	μ/μ_o	O_{II}/O_I	O_{III}/O_I	{[Cd]+[In]}/[O]
S1	CdO-300℃-D1	1.00	0.33	0.18	1.009
S2	CdO-300℃-D120	0.41	0.35	0.30	0.961
S3	CdO-500℃-D120	1.02	0.16	0.25	0.973
S4	CdO-8.3%-300℃-D1	1.00	0.44	0.18	1.003
S5	CdO-8.3%-300℃-D120	0.27	0.38	0.33	1.051
S6	CdO-8.3%-500℃-D120	1.04	0.11	0.33	0.983

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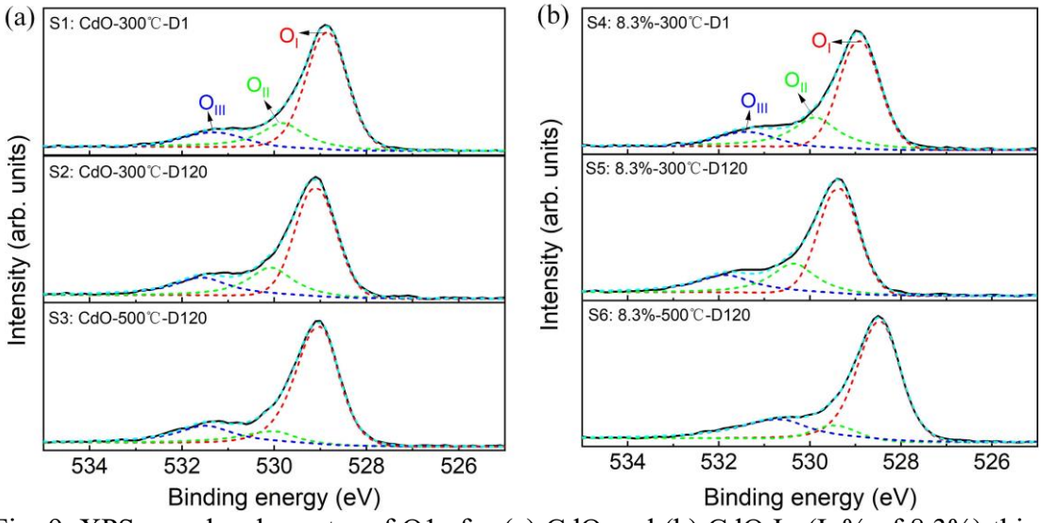


Fig. 9. XPS core-level spectra of O1s for (a) CdO and (b) CdO:In (In% of 8.3%) thin films after PTA at different temperature (300 ºC or 500 ºC) for 10 minutes in Ar, followed by aging in ambient air for either 1 or 120 days.

Be noted that we neglected possible contribution of plasmon satellite for the high binding energy spectral shoulder in the above discussion. For the CdO with electron concentration of ~1.3×10^{20} cm^{-3}, the O_{II} peak in the O 1s spectrum is higher than that of O_I by ~1 eV (see Fig. 9a), which is much larger than the calculated plasmon energy of ~0.34 eV (assuming an effective mass ~0.24 m_0 and high frequency dielectric constant ε_∞ of ~5.4) [52]. This suggests that the O_{II} peaks shown in Fig. 9 may be affected by, but cannot stem from the plasmon satellite. In addition, theoretically the intensity of high binding plasmon satellite peaks in low electron concentration metals should increase with decreasing electron concentration N according to a $N^{-1/3}$ power law [52][53]. However, we observe that the peak area ratios of O_{II}/O_I for samples (e.g., S4, S5) with higher electron concentration N (~12×10^{20} cm^{-3}) is even higher than for samples (e.g., S1, S2) with lower electron concentration of ~1×10^{20} cm^{-3}. Also shown in Table 1, the samples (e.g., S6) with PTA treatments at higher temperature (e.g., 500 ºC) exhibit much smaller O_{II}/O_I ratio than samples (S4 or S5) with PTA treatments at lower temperature (e.g., 300 ºC), even though the former has a lower N than the latter. Since these CdO films are polycrystalline, the presence of substantial defects in the films (especially at the grain boundaries) would inevitably cause significant influence on their XPS spectra. Therefore, we believe that the O_{II} as indicated in the Fig. 9 can be primarily ascribed to the oxygen in the vicinity of oxygen vacancies, though it also includes a plasmon satellite due to the conduction-band plasmon in the accumulation layer.

In general, the optoelectronic properties and environmental stability of polycrystalline TCOs highly depend on the presence of defects. Point defects such as vacancies, interstitials and substitutional impurities are typically far more abundant at grain boundaries compared to the crystal matrix[54][55]. PTA processes can be effective in removing/reducing these defects, especially at high temperatures. Our XPS analysis can help to shed some light on the observed environmental stability behavior of CdO:In thin films in relation to defect evolution at grain boundaries. Our results indicate that unstable CdO:In samples without PTA at high temperature (e.g., 400-500

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ºC) are typically more oxygen deficiency, and most likely with the presence of V_O at grain boundaries. This is illustrated in the schematic diagram shown in Fig. 10(a). A high density of V_O is present at the grain boundary, creating numerous dangling bonds on adjacent Cd atoms. These dangling bonds react with OH^- ions in water moisture (due to electrostatic attraction of OH^- ions by the positive ionized V_O^+ and or V_O^{2+}), resulting in the formation of $Cd(OH)_2$. When free electrons transport through the grain boundaries, they experience strong scattering from these $Cd(OH)_2$ species, leading to a significant reduction in μ. It is important to note that the thermal decomposition of $Cd(OH)_2$ can take place at temperatures ranging from 170 to 300 ºC, where $Cd(OH)_2$ decomposes into CdO and H_2O. Therefore, the degraded CdO:In thin film can be recovered through PTA at relatively low temperature (e.g., ~300 ºC) as observed in our experiment. This recovery is attributed to the thermal decomposition of $Cd(OH)_2$.

Upon thermal annealing at higher temperatures (400~500 ºC), point defects in the materials redistribute. For example, additional cation vacancies may form at the grain surface/boundaries, which is commonly observed in various oxide semiconductors. High-temperature electron paramagnetic resonance (EPR) measurements on ZnO nanoparticles have shown the prominent formation of Zn vacancies at surfaces of ZnO nanoparticles at temperature exceeding 300 ºC [56]. Ab initio calculations suggested that Zn vacancies segregate at the surface of ZnO nanoparticles, forming a Zn free oxo-hydroxide monolayer [57]. In the case of CdO:In thin films, it is plausible that they become cation-deficient after PTA at high temperature, as evidenced by XPS measurements, since the formation energy of V_{Cd} in the highly degenerate CdO (i.e., E_F-VBM>2 eV) is quite low, especially under Cd-poor/O-rich conditions [19]. In fact, high-temperature annealing or thermal calcination has been proven effective in generating a high density of cation vacancies in metal oxides [58][59]. Fig. 10(b) is a schematic representation of a Cd deficient (with a high density of V_{Cd}) grain boundary in CdO:In thin film after PTA at high temperature. These V_{Cd}-induced dangling bonds on O atoms can be readily passivated by available H, forming stable O-H bonds during the high temperature annealing process. Since V_{Cd} acts as a double acceptor in CdO:In, it can be fully passivated by two H atoms. However, partial passivation of V_{Cd} with a single H atom can also occur, resulting in the formation of single acceptors (i.e., $(V_{Cd}\text{-}H)^{-1}$), similar to the H passivation of cation vacancies in other oxide semiconductors [44][60]. First-principle calculations indicated that these hydrogenated cation vacancies (a type of defect complexes) in typical oxide semiconductors like SnO_2, In_2O_3, Ga_2O_3 can be stable up to a temperature exceeding 700 ºC [60]. This suggests that these hydrogenated cation vacancies (i.e., O-H bonds), as shown in Fig. 10(b), can exhibit thermal and chemical stability. Since PTA treatments have negligible influence on the grain size of CdO:In thin films, their possible contribution to eliminating grain boundary defects by forming larger grains is insignificant. In contrast to the $Cd(OH)_2$ species (see Fig. 10a), electron scattering by these hydrogenated cation vacancies is likely much weaker, allowing for the maintenance of high Hall mobility.

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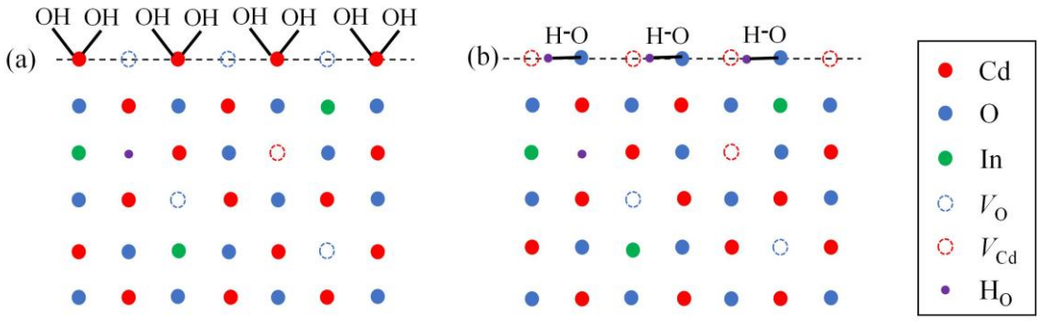


Fig. 10. Schematic illustration of grain boundary (after exposure to moisture) in the CdO:In thin film without (a) and with (b) high temperature PTA. The dashed lines represent the grain boundary. Typical point defects (e.g., V_O, V_{Cd}, In_{Cd}, H_O, V_{Cd}-H) are also shown.

IV. CONCLUSIONS

To conclude, we have investigated the defect-dependent optoelectronic properties of transparent conducting CdO:In thin films with In content ranging from 0 to ~8.3%. The environmental stability of the films was studied by aging them at room temperature in controlled conditions with varying RH from ~0 to ~85%. For as-grown CdO:In thin films, the N increases linearly from ~1×10^{20} cm^{-3} (for CdO) to ~1.2×10^{21} cm^{-3} (with 8.3% In) with increasing In%, while the corresponding μ stays in the range of 80~108 $cm^2V^{-1}s^{-1}$. As a result, a ρ of 5.2×10^{-4}~4.9×10^{-5} Ω cm which is lower than most commercial TCOs can be achieved with CdO:In. After PTA at T_A of 400~500 ºC in either Ar or ambient air for 10 minutes, the CdO:In thin films with high In% (>6%) show notable reduction in N due to dopant segregation at high doping concentration. The E_G^{opt} of these CdO:In thin films is found to be in the range of 2.4~3.1 eV, depending on their N. SE analysis indicates that these highly degenerate CdO:In thin films can exhibit superior transparency (>80%) in the NIR region (>1000 nm) compared to the typical ITO or AZO thin film with similar sheet resistance. Therefore, these CdO:In thin films hold great promise for optoelectronic devices utilizing photons from the UV to NIR spectral range (~400 to 1400 nm), such as full-spectrum solar cells or infrared detectors. However, these as-grown CdO:In thin films exhibit degradation in the μ when exposed to moisture. Such environmental instability is not observed when the films were subjected to PTA treatments at higher T_A of 400~500 ºC in Ar or air for ~10 minutes. XPS results reveal that unstable CdO:In samples (without PTA treatment at high T_A) are typically oxygen-deficient (with a higher density of V_O at the grain boundaries), and the Cd dangling bonds associated with V_O tend to attract OH^- ions in water moisture and form $Cd(OH)_2$, leading to the degradation in the mobility. On the other hand, stable CdO:In samples (with PTA treatment at high T_A) are cation-deficient (with a higher density of V_{Cd} at grain boundaries), and H could passivate the V_{Cd}-related defects at grain boundaries during high-temperature annealing process. Overall, this study provides a comprehensive understanding of the instability mechanisms and defect passivation in high mobility transparent conducting CdO:In thin films, which are essential for achieving outstanding optoelectronic performance in a wide spectral window.

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DATA AVILABILITY

The data that supports the findings of this study are available from the corresponding author upon reasonable request.

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